LOCALITY OF COHOMOLOGY: A COHOMOLOGY CLASS ON A SITE DIES ON A COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed. The author writes 'is exact in' where 'is exact at' is meant; this is reproduced verbatim and not corrected.

Editorial note. The preamble is the chapter's own native preamble excerpt used for this source: the Xy-pic setup at lines 14-18 and the theorem declarations with the mathematical macros at lines 154-217 of sources/stacks-etale-cohomology/preamble.tex. The quoted class line and the xr-hyper externaldocument block are not included because the assembler supplies a plain amsart wrapper; no macro was invented or redefined.

Editorial note. The cross-chapter pointer is located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim line carrying its canonical tag.

Allowed standard prerequisites (editorial).

- Cohomology of an abelian sheaf on a site computed from an injective resolution. Standard in this chapter: H^p(U, F) is the p-th cohomology of the complex of sections of an injective resolution F -> I^bullet. Supplies the representative of the class and the fact that restriction is computed on representatives; the covering argument is the counted content.
- Exactness of the resolution in positive degrees means local triviality of sections of the kernel. Standard sheaf theory: I^{p-1} -> I^p -> I^{p+1} exact in Ab(C) means a section over U killed by d becomes a coboundary after restriction to a covering. Supplies the existence of the local primitives xi_i; identifying them and concluding the vanishing is inside the counted proof.

Source reference. Original lemma 01FW: sources/stacks-chow/tags.txt, source lines 1885–1885. Cross-chapter 'for more details' pointer at proof line 2276, invoked after the argument is complete; the resolution, the local primitives and the vanishing are all retained verbatim in the counted span, so this pointer replaces nothing.

Lemma 0.1 (Locality of cohomology)**.** *Let $\mathcal{C}$ be a site, $\mathcal{F}$ an abelian sheaf on $\mathcal{C}$, U an object of $\mathcal{C}$, $p > 0$ an integer and $\xi \in H^p(U, \mathcal{F})$. Then there exists a covering $\mathcal{U} = \{U_i \to U\}_{i \in I}$ of U in $\mathcal{C}$ such that $\xi|_{U_i} = 0$ for all $i \in I$.*

Proof. Choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$. Then ξ is represented by a cocycle $\tilde{\xi} \in \mathcal{I}^p(U)$ with $d^p(\tilde{\xi}) = 0$. By assumption, the sequence $\mathcal{I}^{p-1} \to \mathcal{I}^p \to \mathcal{I}^{p+1}$ in exact in $Ab(\mathcal{C})$, which means that there exists a covering $\mathcal{U} = \{U_i \to U\}_{i \in I}$ such that $\tilde{\xi}|_{U_i} = d^{p-1}(\xi_i)$ for some $\xi_i \in \mathcal{I}^{p-1}(U_i)$. Since the cohomology class $\xi|_{U_i}$ is represented by the cocycle $\tilde{\xi}|_{U_i}$ which is a coboundary, it vanishes. For more details see Cohomology on Sites, Lemma 01FW. □

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